COMPASS_TEMPEST_SGW_2024: September

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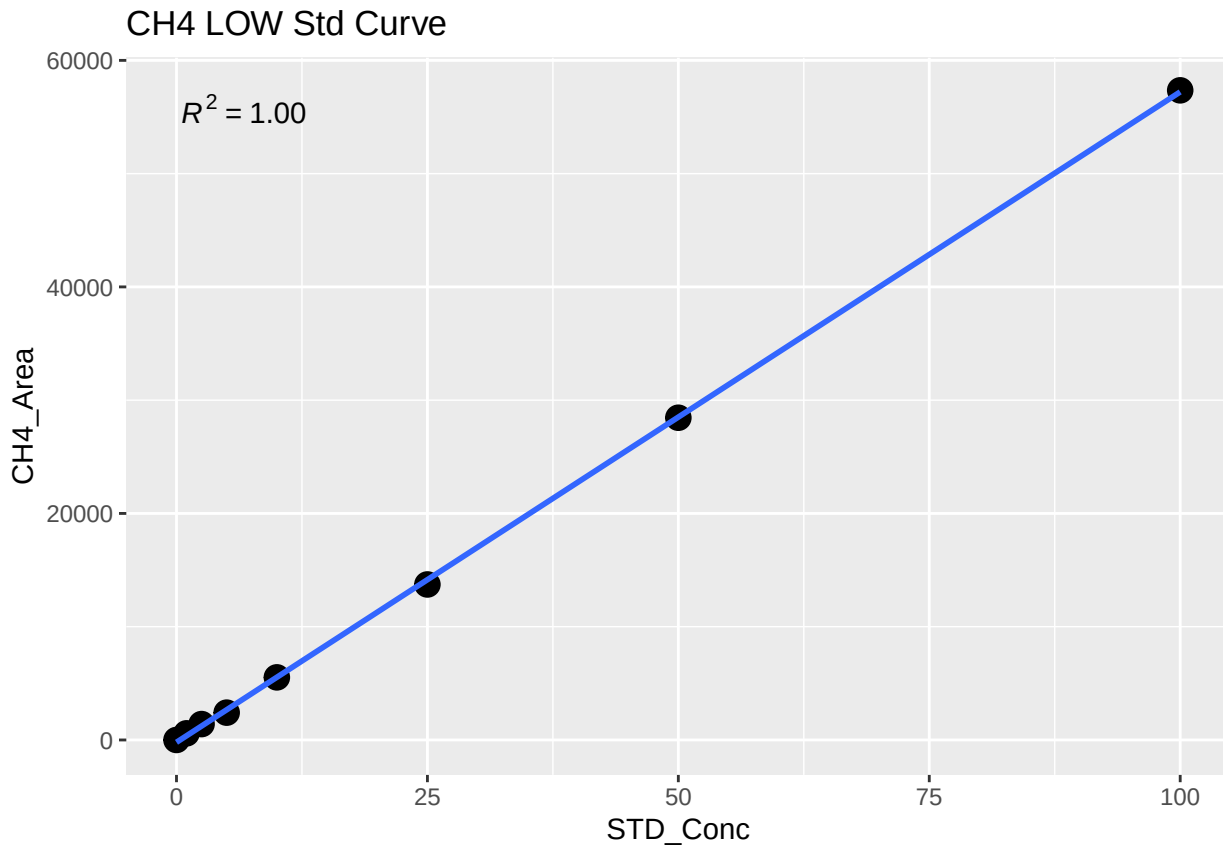
##Set Up

Read in first data file and assess standard curves

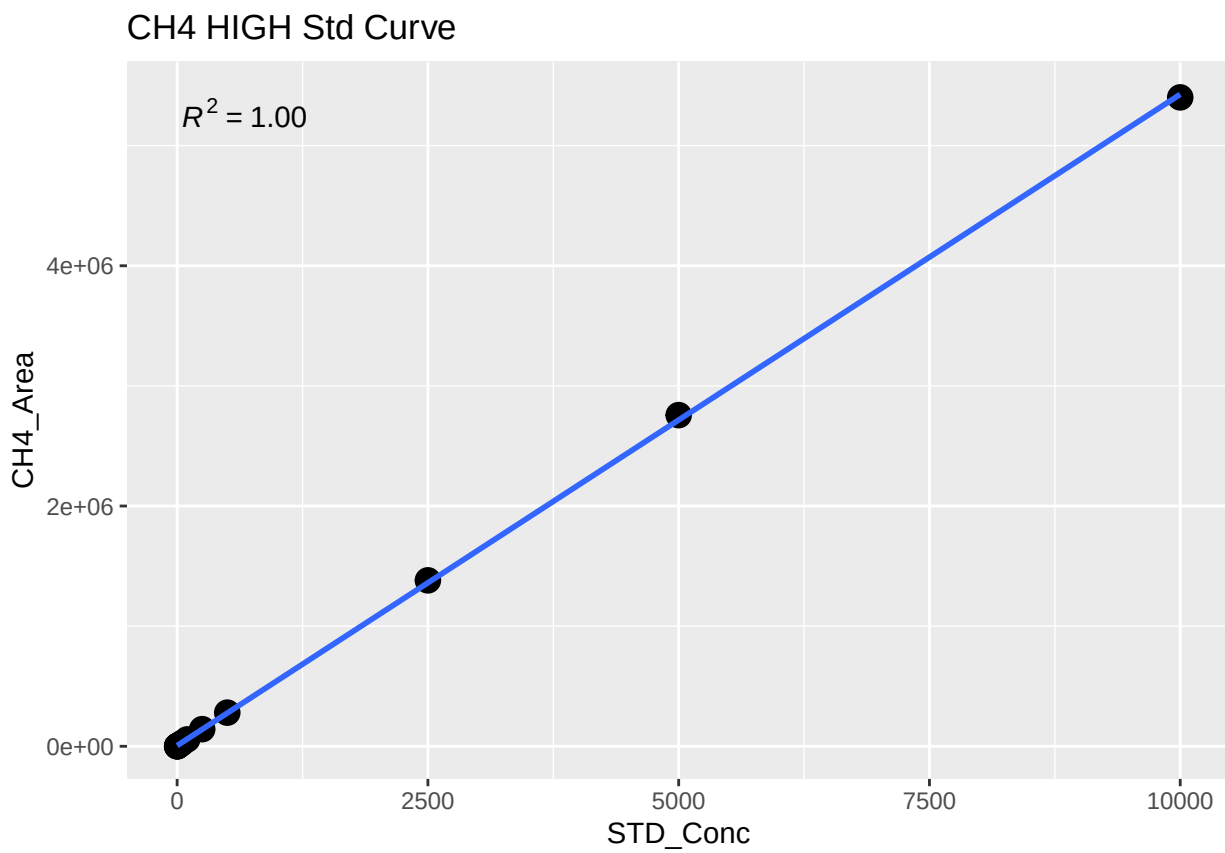
```
## Standard curve run with this run had standards that were not prepared properly (over pressurized)
## We have brought in the curve run most recently to this run that was prepared correctly for analysis
```

```
##     Machine       User Run_Date Sample_Type Sample1 Sample_Year Sample_Month
## 1 Varian GC A. Stearns 20240910       Blank   Blank        2024         <NA>
## 2 Varian GC A. Stearns 20240910     STD_CH4 STD_CH4        2024         <NA>
## 3 Varian GC A. Stearns 20240910     STD_CO2 STD_CO2        2024         <NA>
## 4 Varian GC A. Stearns 20240910     STD_CH4 STD_CH4        2024         <NA>
## 5 Varian GC A. Stearns 20240910     STD_CH4 STD_CH4        2024         <NA>
## 6 Varian GC A. Stearns 20240910     STD_CH4 STD_CH4        2024         <NA>
##   Sample_Day Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth
## 1         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 2         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 3         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 4         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 5         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 6         NA          NA      <NA>   G        <NA>        <NA>         <NA>
##              Sample_ID Dilution_Factor STD_Conc CO2_Area CH4_Area Field_Notes
## 1              Blank_0               1      0.0        0     2517          NA
## 2 Blank_0_repeatforCH4               1      0.0        0     2517          NA
## 3 Blank_0_repeatforCO2               1      0.0        0     2517          NA
## 4         STD_1ppm_CH4               1      1.0    24162      518          NA
## 5       STD_2.5ppm_CH4               1      2.5    67043     1260          NA
## 6         STD_5ppm_CH4               1      5.0   131230     2423          NA
##   Lab_Notes
## 1
## 2
## 3
## 4
## 5
## 6
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

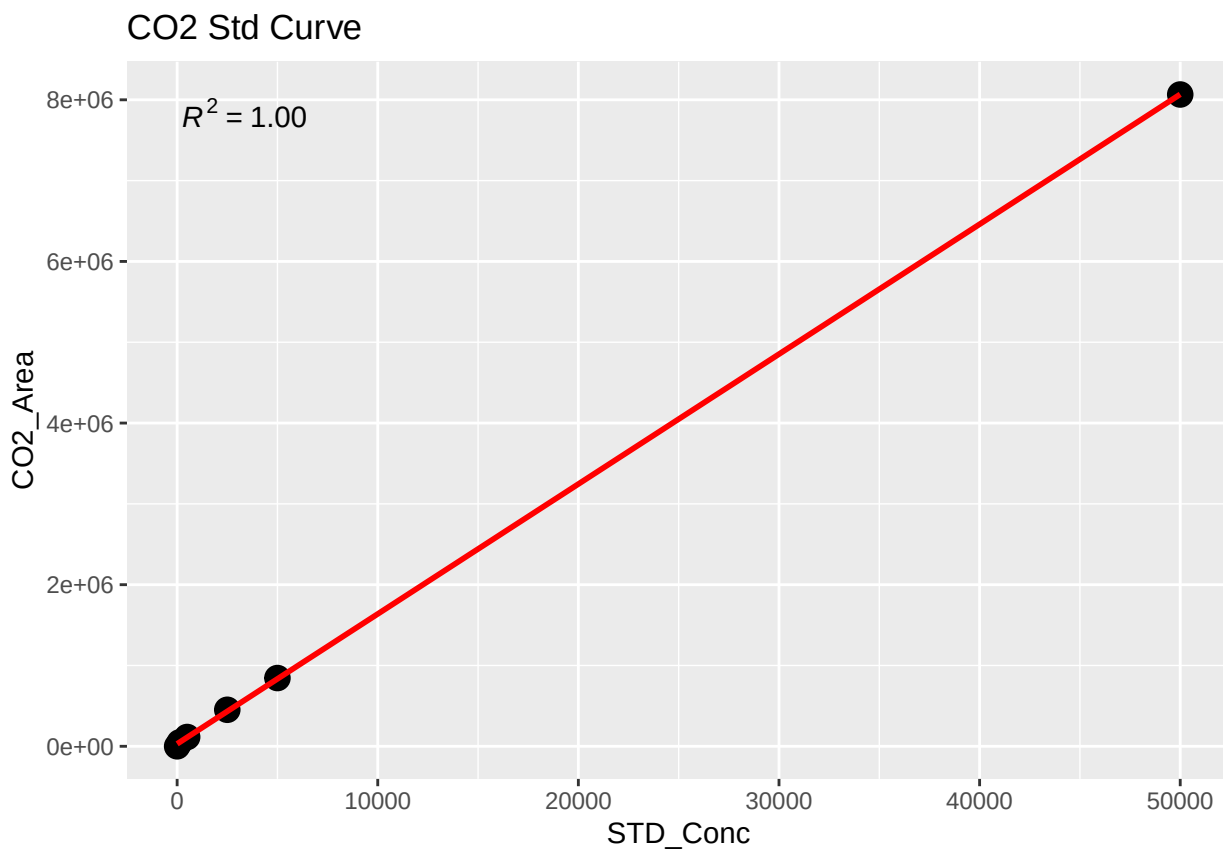



```
##
## Call:
## lm(formula = stds_ch4_low$CH4_Area ~ stds_ch4_low$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -420.05 -104.77   65.47  181.95  217.14
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -206.352    111.046  -1.858    0.112
## stds_ch4_low$STD_Conc  574.216      2.728 210.502 7.76e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 252.6 on 6 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9998
## F-statistic: 4.431e+04 on 1 and 6 DF,  p-value: 7.756e-13

## `geom_smooth()` using formula = 'y ~ x'
```

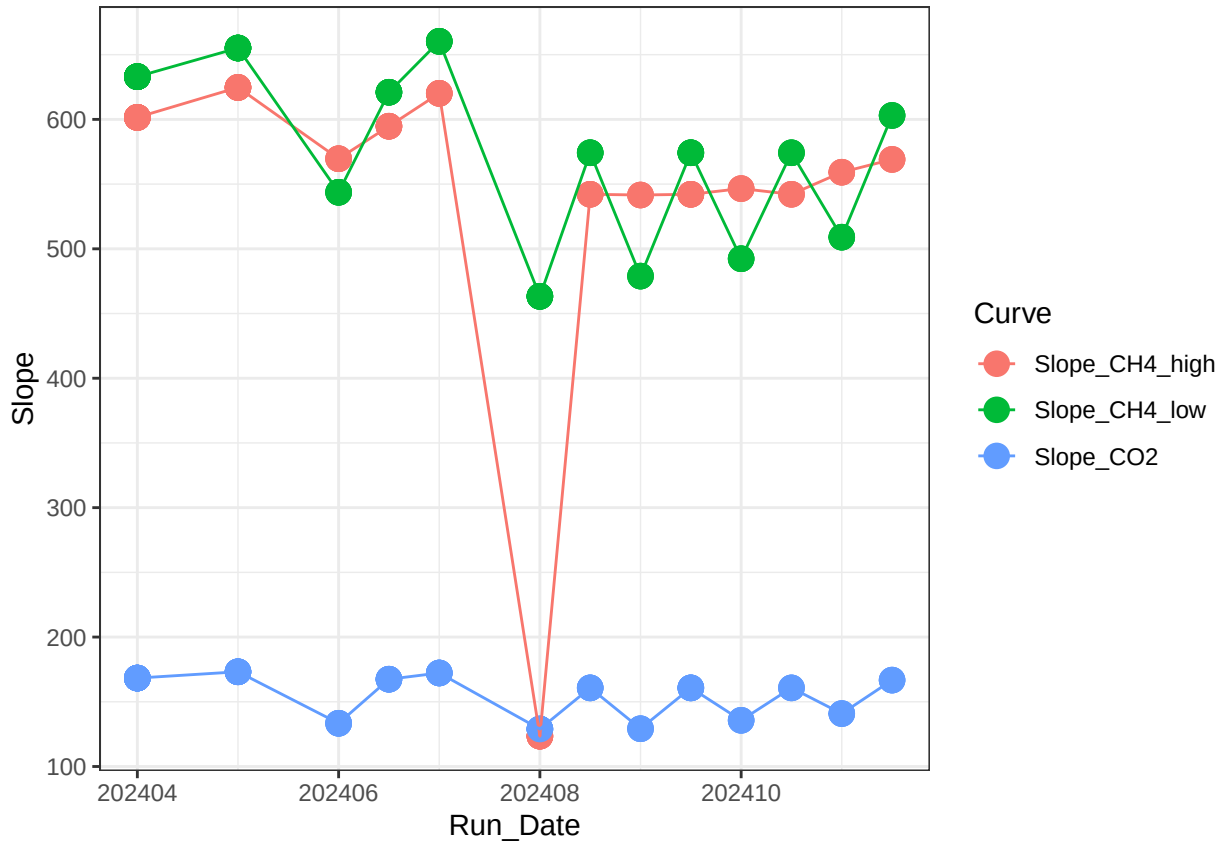



```
##
## Call:
## lm(formula = stds_ch4$CH4_Area ~ stds_ch4$STD_Conc)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -25417  -5566  -5433   1511  40362
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        5608.636   5046.024   1.111     0.29
## stds_ch4$STD_Conc   542.125      1.586 341.796   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16280 on 11 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 1.168e+05 on 1 and 11 DF,  p-value: < 2.2e-16
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = stds_co2$CO2_Area ~ stds_co2$STD_Conc)
##
## Residuals:
##      1      2      3      4      5      6
## -30347  -2119   4272  19689  10586  -2080
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.035e+04  8.808e+03   3.445   0.0262 *
## stds_co2$STD_Conc 1.607e+02  4.288e-01 374.873 3.04e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19020 on 4 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 1.405e+05 on 1 and 4 DF,  p-value: 3.038e-10

##   X          Curve        R2    Slope  Intercept Run_Date
## 1 1  Slope_CH4_low 0.9996407 633.0314  -320.6965   202404
## 2 2 Slope_CH4_high 0.9993443 601.5512 14817.1191   202404
## 3 3      Slope_CO2 0.9999907 168.3200 10075.2183   202404
## 4 4  Slope_CH4_low 0.9996407 633.0314  -320.6965   202404
## 5 5 Slope_CH4_high 0.9993443 601.5512 14817.1191   202404
## 6 6      Slope_CO2 0.9999907 168.3200 10075.2183   202404
```



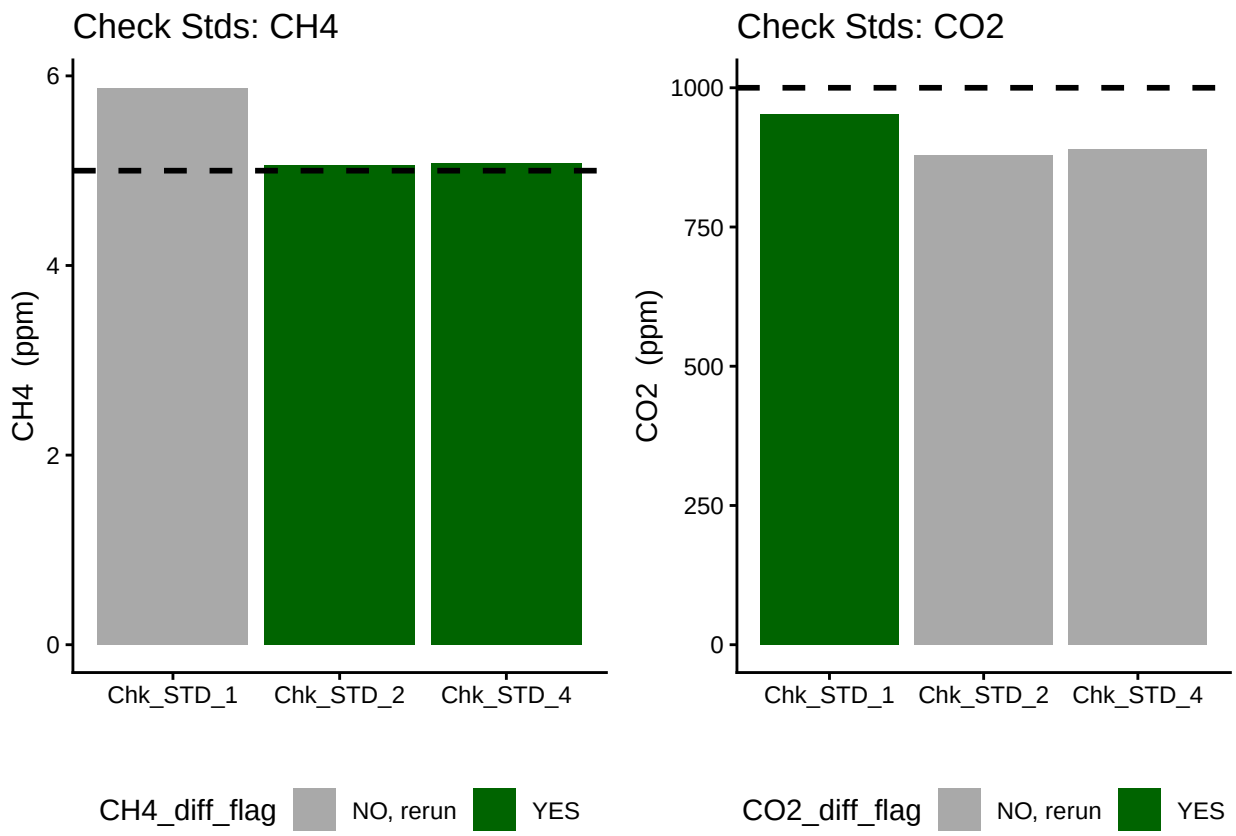
Now calculate the CH4 & CO2 concentrations in ppm

```
#head(raw)

#pull out methane standards
Samples <-  raw %>%
  filter(!str_detect(Sample_Type, "STD_CH4")) %>%
  filter(!str_detect(Sample_Type, "STD_CO2")) %>%
  filter(!str_detect(Sample_Type, "Blank")) %>%
  filter(!str_detect(Sample_Type, "Chk_STD")) %>%
   filter(!str_detect(Sample_Type, "CHKSTD")) %>%
  filter(!str_detect(Sample_Type, "NA"))
#head(Samples)

#Now flag any areas that are above the 100ppm area for CH4
Samples$CH4_Curve <- ifelse(Samples$CH4_Area >71000, "High", "Low")
#head(Samples)

#Calculate CH4 concentrations in ppm
Samples$CH4_Conc_ppm <- ifelse(Samples$CH4_Area >71000, (Samples$CH4_Area-Slope_CH4_high$Intercept)/Slo

#Calculate CO2 concentrations in ppm
Samples$CO2_Conc_ppm <- ((Samples$CO2_Area-Slope_CO2$Intercept)/Slope_CO2$Slope)
```

```r
#head(Samples)

##########make flags for any dilutions needed
#highest CH4 standard = 10000
#highest CO2 standard = 50000

Samples$CH4_Flag <- ifelse(Samples$CH4_Conc_ppm >10000, "Needs Dilution", "Within Range")
Samples$CO2_Flag <- ifelse(Samples$CO2_Conc_ppm >50000, "Needs Dilution", "Within Range")
#head(Samples)
```

Check the Check Standards

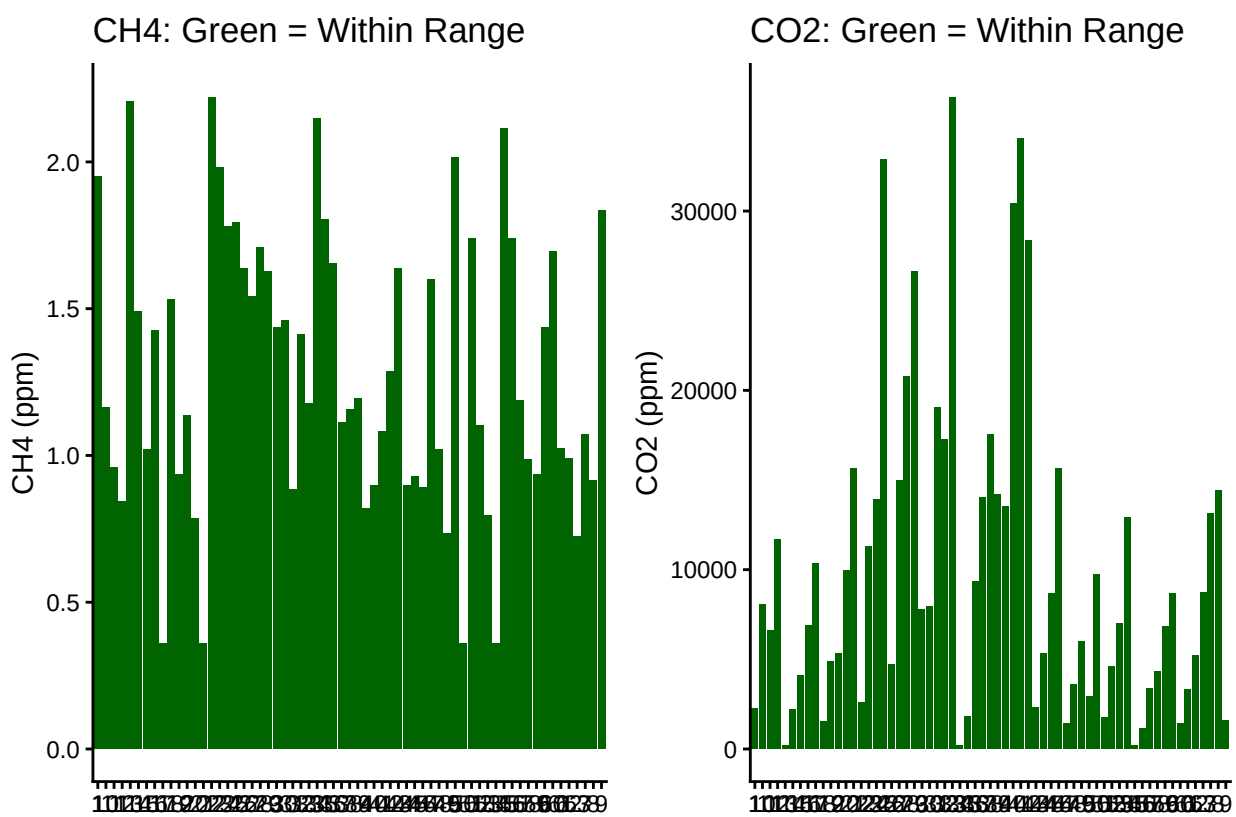
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



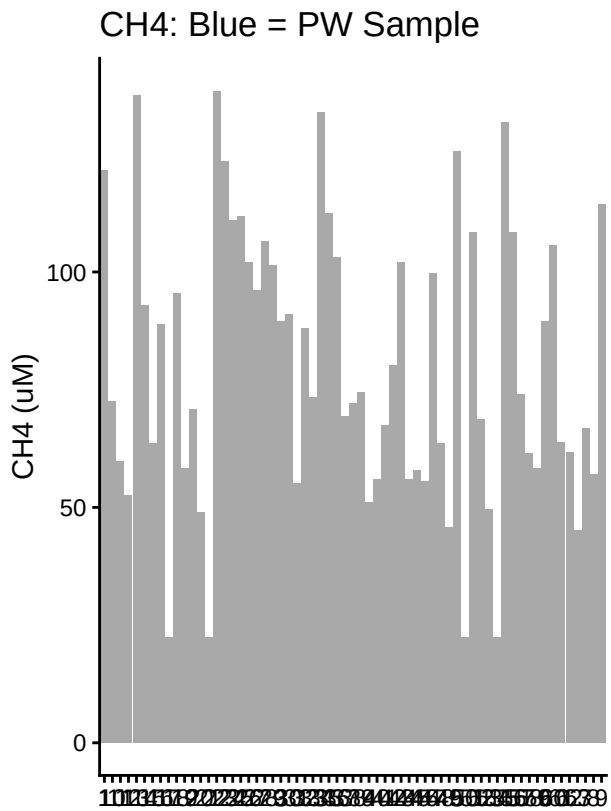
Dilution correct samples

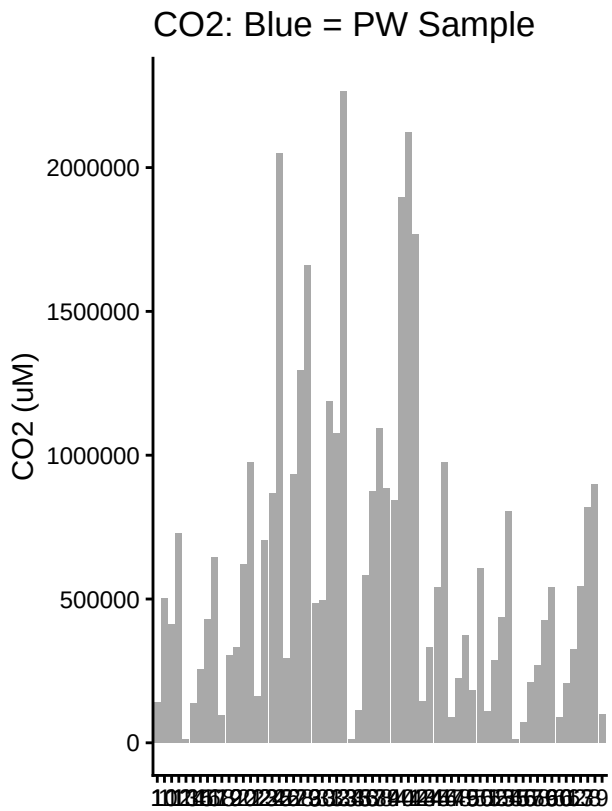
```r
#multiply the concentration by the dilution factor
Samples$CH4_Conc_ppm_dilcorr <- (Samples$CH4_Conc_ppm * Samples$Dilution_Factor)
```

```r
Samples$CO2_Conc_ppm_dilcorr <- (Samples$CO2_Conc_ppm * Samples$Dilution_Factor)

#check results
#head(Samples)

#quick first look at the samples
ch4_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CH4_Conc_ppm, fill=CH4_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("darkgreen","red"))+
        #scale_fill_gradient2(low='red', mid='white', high='blue', space='Lab') +
        theme_classic() + labs(x= " ", y="CH4 (ppm)", title="CH4: Green = Within Range") +
        theme(legend.position="none")


co2_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CO2_Conc_ppm, fill=CO2_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c( "darkgreen", "red"))+
        #scale_fill_gradient2(low='red', mid='white', high='blue', space='Lab') +
        theme_classic() + labs(x= " ", y="CO2 (ppm)", title="CO2: Green = Within Range") +
        theme(legend.position="none")

ggarrange(ch4_samples, co2_samples, nrow=1, ncol=2)
```



If samples are water calculate gas in water





Write out processed data & slopes

```
#check results
#head(Samples)

#pull out what we need
Samples1 <- Samples[ ,c(1:3,6:16,20, 24:30)]
#head(Samples1)

#change the column names as needed
#colnames(samples1) <- c("Machine", "User", "Run_Date" , "Sample_Year" ,  "Sample_Month" , "Sample_Day"
 #                       "Sample_Time" , "Time_Zone" , "G_W" , "Sample_Plot","Sample_Grid","Sample_Dept
  #                      "Sample_ID" ,  "Dilution_Factor", "Notes" ,  "CH4_Flag", "CO2_Flag" ,
   #                     "CH4_Conc_ppm_dilcorr", "CO2_Conc_ppm_dilcorr")

write.csv(Samples1, "2024 Processed Data/COMPASS_TEMPEST_SGW_202409_Processed.csv")
```

#end